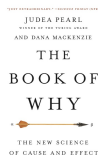
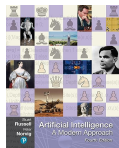


Lesson 08.2: Causal Inference

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References:

- AIMA (Artificial Intelligence: a Modern Approach)
- Pearl et al., The Book of Why, 2017



- ***Causal Networks***
 - Causal DAGs
 - Structural Causal Model
 - Variables
 - Type of Variables in Causal AI
 - Types of Paths in Causal AI

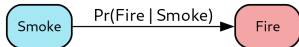
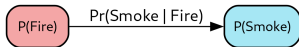
- Causal Networks
 - *Causal DAGs*
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(Non-Causal) Bayesian Networks

- **Bayesian networks** represent a joint distribution function
 - The direction of the arrow represent *conditional dependence* (not causality)
 - $A \rightarrow B$ requires to estimate $\Pr(A|B)$
- **Many possible Bayesian networks** with same nodes, different edges to explain the same phenomenon

- **Example**

- A Bayesian network with *Fire* and *Smoke*, which are dependent
- $Fire \rightarrow Smoke$
 - * Need $\Pr(Fire)$ and $\Pr(Smoke|Fire)$ to compute $\Pr(Fire, Smoke)$
- $Smoke \rightarrow Fire$
 - * Need $\Pr(Smoke)$ and $\Pr(Fire|Smoke)$



- **Different Bayesian networks:**
 - Are equivalent and convey the same information
 - Have different difficulties to be estimated

Causal (Bayesian) Networks

- **Causal networks are Bayesian networks with only causal edges**
 - Use judgment based on nature instead of just statistics
 - E.g., you need to go from
 - * “Are random variables *Smoke* and *Fire* correlated?” to
 - * “What causes what, *Smoke* or *Fire*?”
- **"Dependency in nature"** is like assignment in programming
 - E.g., nature assigns *Smoke* based on *Fire*:
 - *  *Smoke* := $f(\textit{Fire})$
 - *  *Fire* := $f(\textit{Smoke})$
- **Structural equations** describe “assignment mechanism” in causal graphs

$$X_i := f(X_j) \iff X_j \rightarrow X_i$$

Causal DAG

- **Causal DAG**

- *Directed*: Arrows show cause $\rightarrow$ effect
- *Acyclic*: No feedback loops
 - * Causal relationships assume temporal order: cause before effect
 - * A cycle implies a variable is both cause and effect of itself

- **Benefits**

- DAGs makes explicit *causal* links
- Support explainable AI models
- Stability in conditional probability estimation
- Reason about interventions and counterfactuals

- **Limitations**

- Requires domain knowledge for structure
- Assumes all relevant variables included (no hidden confounders)

Causal Edges are Stable

- **Causal edges reflect stable relationship**
 - *Mechanistic stability*
 - * Causal relationships show system function, not just behavior in one dataset
 - * E.g., “*Temperature* $\rightarrow$ *ice melting rate*” holds true in Alaska and Arizona
 - *Invariance under interventions*
 - * If X causes Y , intervening on X affects Y consistently, despite confounders or context changes
 - *Easier estimation through causal modeling*
 - * Identifying causal direction focuses estimation on effect size (e.g., regression of Y on X under intervention)
- **Example:** study *Exercise* $\rightarrow$ *Health*:
 - Correlation may differ in young or elderly populations
 - Causal effect remains stable, as physiological mechanism doesn't change

Causal DAG: Example

- **Explanatory variables**

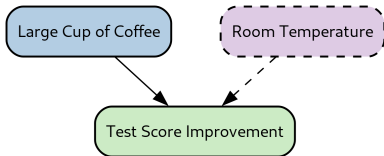
- You can manipulate or observe when changes are applied
- E.g., *“does a large cup of coffee before an exam help with a test?”*

- **Outcome variables**

- Result of the action
- E.g., *“by how much did the score test improve?”*

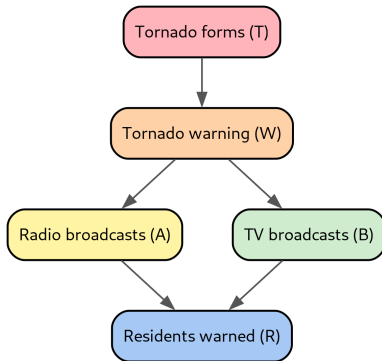
- **Unobserved variables**

- Not seen or more difficult to account
- E.g., *“temperature of the room makes students sleepy and less alert”*



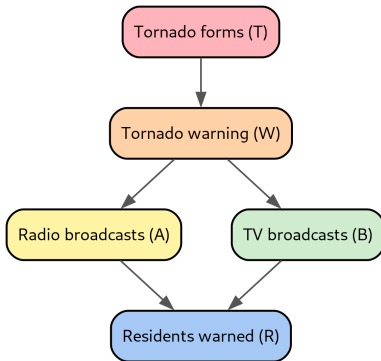
Example: Tornado Warning

- **Example:** Severe weather alert to the public
- **Causal diagram:**
 - T : Tornado forms
 - W : National Weather Service issues tornado warning
 - A, B : Radio and TV broadcast emergency
 - R : Residents warned
- **Assumptions**
 - Channels broadcast only on official order
 - Delivery systems never fail
 - If either channel activates, residents warned



Tornado Warning: Level 1 (Association)

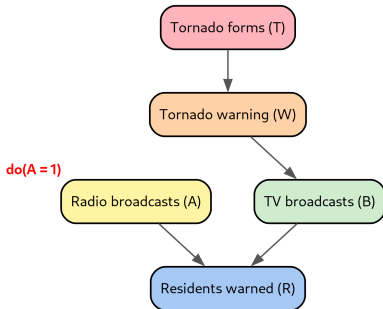
- **Use causal graph to understand how one fact informs another**
 - Use predicate logic to answer
- **If residents are warned, was a warning issued?**
 - Does $R \implies W$?
 - Yes
- **If radio *A* broadcast, did TV *B* also broadcast?**
 - Yes
 - True even though *A* doesn't cause *B*
 - Correlated through common cause *W* (confounder)



Tornado Warning: Level 2 (Intervention)

- **If you make radio broadcast, will residents be warned?**

- Yes
- Breaks “rules of nature” as A broadcasts only if W is issued
 - * Like removing edge $W \rightarrow A$ in graph
 - * Set A to *true*:
 $do(A = 1)$
- Expect B did *not* broadcast
 - * Forcing A shouldn't affect B (no $A \rightarrow B$ edge)



- **Key difference between "seeing" and "doing"**

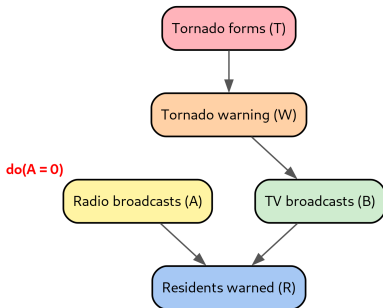
- If see A broadcast, conclude B broadcast too
- If *make* A broadcast, expect B didn't

- **Observational data wouldn't reveal causal structure**

- Either all variables T, W, A, B, R are true or all are false
- Correlations alone are uninformative

Tornado Warning: Level 3 (Counterfactual)

- **Assume that residents are warned $R = 1$**
 - You know A and B broadcast because
 - ... a warning W was issued
 - ... after a tornado formed T
- **Would residents be warned if A had not broadcast?**
 - Remove the edge into A
 - Set $do(A = 0)$
 - TV B still broadcasts (due to W), and $R = (A \vee B)$ remains true
 - Yes, $R = 1$



- Causal Networks
 - Causal DAGs
 - ***Structural Causal Model***
 - Variables
 - Type of Variables in Causal AI
 - Types of Paths in Causal AI

Structural Causal Model

- A **Structural Causal Model** (SCM) translates a causal DAG into mathematical equations to define how variables interact
- **Structure of SCMs**
 - Variables $X_1, X_2, \dots, X_n$ represent quantities in the system
 - *Equations* model each variable as a function of its direct causes
 - Formally, X_i is modeled as:

$$X_i = f_i(\text{Parents}(X_i), \varepsilon_i)$$

where:

- * $\text{Parents}(X_i)$ are direct causes of X_i
- * ε_i is an exogenous (external, unobserved) noise term
- **Properties**
 - Same properties of causal networks
 - * Explain causal relationships between variables
 - * Provide a foundation for causal reasoning and simulation
 - * ...
 - Quantify effect

– Used in econometrics and genetics for a long time (even before theory of

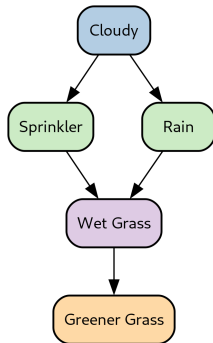
Structural Causal Model: Sprinkler Example

- **Structural equations** for this causal DAG:

$$\begin{cases} C := f_C(\varepsilon_C) \\ R := f_R(C, \varepsilon_R) \\ S := f_S(C, \varepsilon_S) \\ W := f_W(R, S, \varepsilon_W) \\ G := f_G(W, \varepsilon_G) \end{cases}$$

- **Unmodeled variables** ε_x represent error terms
 - E.g., ε_W is another source of wetness (e.g., *MorningDew*) besides *Sprinkler* and *Rain*
 - Assume unmodeled variables are exogenous, independent, with a certain distribution (prior)
- Express **joint distribution** of all variables as a product of conditional distributions using causal DAG topology:

$$\Pr(C, R, S, W, G) = \Pr(W|R, S) \Pr(G|W) \Pr(S|C) \Pr(R|C) \Pr(C)$$

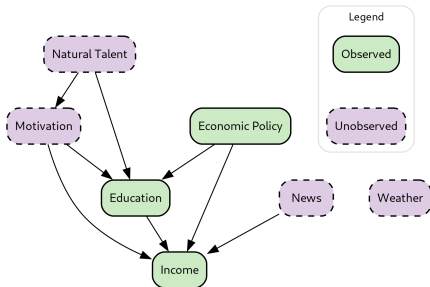


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Observed Vs. Unobserved Variables

■ Observed variables

- Aka “measurable” or “visible”
- Variables directly measured or collected in a dataset
- E.g.,
 - * Education
 - * Income



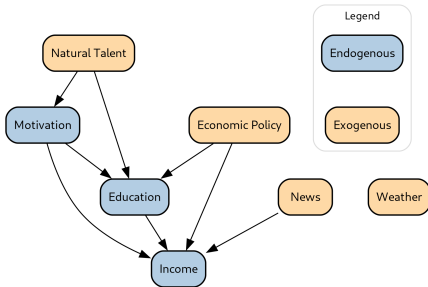
■ Unobserved variables

- Aka “latent” or “hidden”
- Exist but not measured or included in data
- E.g.,
 - * Natural talent
 - * Motivation
- Ignoring unobserved variables leads to incorrect conclusions
 - * E.g., $IceCreamSales \leftarrow Temperature \rightarrow DrowningRates$

Endogenous Vs. Exogenous Variables

■ Endogenous variables

- Values determined *within* the model
 - * Dependent on other variables in the system
- Represent system's internal behavior and outcomes
- E.g.,
 - * Motivation
 - * Income



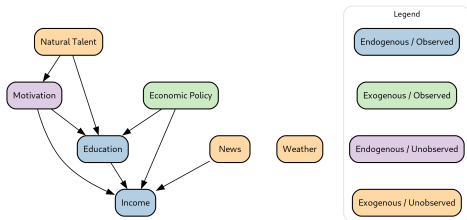
■ Exogenous variables

- Originate *outside* the system being modeled
 - * Not caused by other variables in the model
- Represent background conditions or external shocks
- E.g.,
 - * Natural talent
 - * Economic policy
 - * Weather

Endo / Exogenous, Observed / Unobserved Vars

■ Typically

- *Exogenous / unobserved variables*: capture randomness or unknown external factors
- *Exogenous / observed variables*: potential intervention factors
- *Endogenous / observed variables*: focus for prediction and intervention



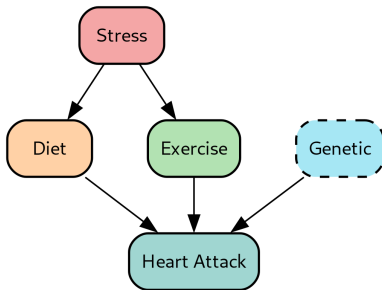
Variable Type	Observability	Example
Endogenous	Observed	Income
Exogenous	Observed	Education
Endogenous	Unobserved	Motivation
Exogenous	Unobserved	Natural Talent

Building a Causal DAG

- **Causal models** visually represent complex environments and relationships
 - Nodes are like “nouns”:
 - * E.g., “price”, “sales”, “revenue”, “birth weight”, “gestation period”
 - * Variables can be endogenous/exogenous and observed/unobserved
 - Relationships between variables are “verbs”:
 - * Parents, children (direct relationships)
 - * Descendants, ancestors (along the path)
 - * Neighbors
- **Modeling as a Communication Tool:**
 - Shared language bridges gaps between technical and non-technical team members
- **Iterative Refinement:**
 - Continuously update models with new variables and insights

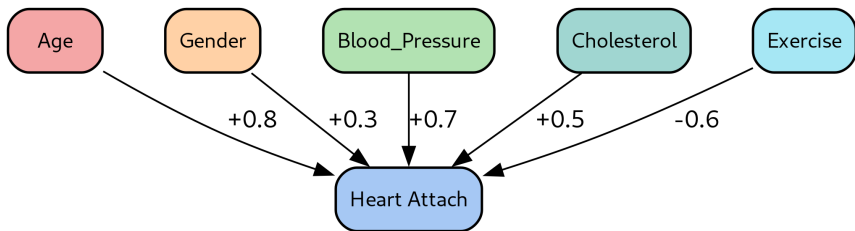
Heart Attack: Example

- Research question: *What's the relationship between stress and heart attacks?*
- Build a causal DAG**
 - *Stress* is the treatment
 - *Heart attack* is the outcome
 - Stress is *not* a direct cause of heart attack
 - * E.g., a stressed person tend to have poor eating habits and tends not to exercise
 - *Genetics* is unobserved



Weights

- Assign weights to paths to represent causal strength
- Sign indicates direction



- **How to estimate sign and weight**
 - Estimate using correlation
 - Use priors and then estimate using Bayesian approach

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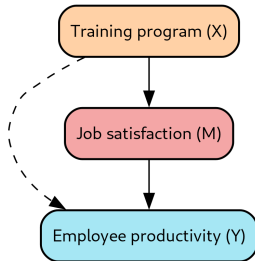
Mediator Variable

- A **mediator variable** M
 - Is an intermediate variable that *transmits* the causal effect from X (treatment) to Y (outcome)
 - Lies **on the causal path** between X and Y
 - Captures the **mechanism or process** through which X influences Y



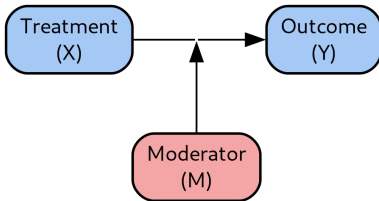
Mediator Variable: Example

- **Research question:** “Does a training program increase employee productivity?”
- Causal effect may be indirect, through a **mediator**
 - Training might not immediately boost productivity
 - Could enhance job satisfaction, raising productivity
- **Causal interpretation**
 - X : Training Program (cause)
 - M : Job Satisfaction (mediator)
 - Y : Employee Productivity (effect)
 - Path: $X \rightarrow M \rightarrow Y$
- **Direct vs. Indirect effects**
 - *Indirect effect:* X affects Y through M
 - *Direct effect:* X affects Y not through M
 - Controlling for M separates effects, clarifying training impact



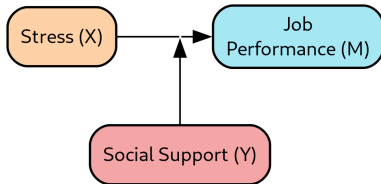
Moderator Variable

- A **moderator variable** M
 - Changes the *strength* or *direction* of the relationship between an independent variable (X) and a dependent variable (Y)
 - Is not part of the causal chain but conditions the relationship



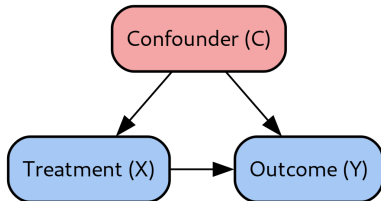
Moderator Variable: Example

- **Research question:** “Study relationship between stress and job performance”
- **Social support** *M* as a moderator
 - High social support weakens stress’s negative effect on performance
 - Low social support strengthens stress’s negative effect on performance



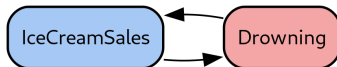
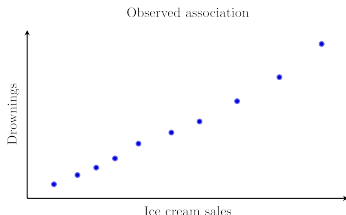
Confounder Variable

- A **confounder** C
 - Affects both treatment (cause) and outcome (effect)
 - Creates misleading association, if not controlled



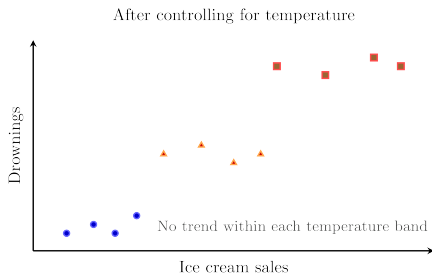
Confounder Variable: Example

- *IceCreamSales* and *Drowning* **move together**
 - Correlation-based model claims association
 - Is it true?
 - You can always find an explanation (e.g., from ChatGPT)
 - * Eating ice cream may distract children or guardians:
 - * Cold food shock reflex causes hyperventilation in water
 - * Sugar spike → hyperactivity near pools
- **How to use this relationship?**
 - Ban ice cream to prevent drowning?
 - Ice cream maker increase drowning to boost sales?

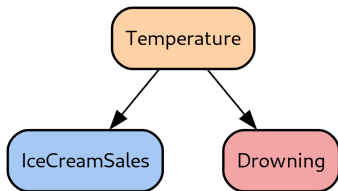


Confounder Variable: Example

- In reality, **no cause-effect** between *IceCreamSales* and *Drowning*
 - *Temperature* is a confounder
- In fact, when control for temperature (in regression or intervention), association disappears

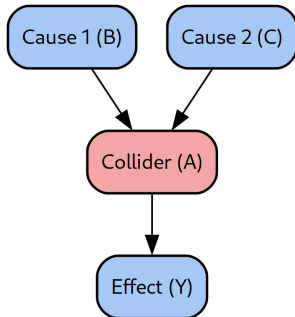


No relationship once temperature is controlled



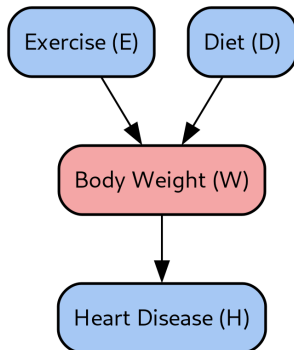
Collider

- A **collider** A
 - Is a variable influenced by multiple variables B , C
 - Complicates understanding relationships between variables B , C and those it influences, Y



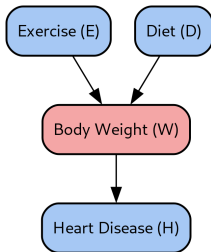
Collider: Examples

- Study the relationship between *Exercise* and *HeartDisease*
 - *Diet* and *Exercise* influence *BodyWeight*
 - *BodyWeight* influences *HeartDisease*
 - *BodyWeight* is a collider



Collider Bias

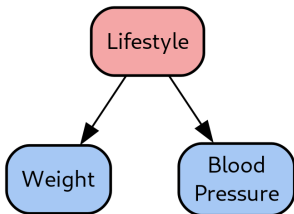
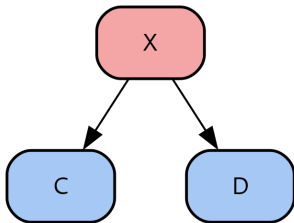
- Aka “Berkson’s paradox”
- **Conditioning on a collider** can introduce a spurious association between its parents by *“opening a path that is blocked”*
- **Example**
 - Diet (D)
 - Exercise (E)
 - BodyWeight (W)
 - HeartDisease (D)
- **Without conditioning on W**
 - E and D are independent
 - * E.g., knowing exercise level E doesn’t inform about diet D, and vice versa
 - Collider W blocks association between E and D
- **After conditioning on W**
 - E.g., individuals with specific body weight
 - Introduce dependency between E and D
 - With W fixed, changes in E balanced by changes in D, inducing spurious correlation between E and D



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Fork Structure

- A **fork** $D \leftarrow X \rightarrow C$ occurs when a single variable causally influences two or more variables
 - X is a **confounder** (common cause) of C and D
 - Forks induce statistical dependence between C and D even if C and D are not causally linked
- **Conditioning** on X *blocks the path* and removes spurious correlation
- **Example**
 - *Lifestyle* is a confounder that affects both *Weight* and *BloodPressure*
 - These outcomes may appear correlated due to shared cause



Inverted Fork

- An **inverted fork** occurs when two or more arrows converge on a common node
 - **Colliders** block associations unless the collider or its descendants are conditioned on
- **Conditioning on a collider** opens a path, inducing spurious correlations
 - This is the basis of selection bias
- **Example**
 - Sales influenced by multiple independent causes
 - *MarketingSpend* and *ProductQuality* both influence *Sales*
 - Conditioning on *Sales* can induce false dependence between *MarketingSpend* and *ProductQuality*

